

# A Model of Visually-Guided Smooth Pursuit Eye Movements Based on Behavioral Observations

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**Abstract.** We report a model that reproduces many of the behavioral properties of smooth pursuit eye movements. The model is a negative-feedback system that uses three parallel visual motion pathways to drive pursuit. The three visual pathways process image motion, defined as target motion with respect to the moving eye, and provide signals related to image velocity, image acceleration, and a transient that occurs at the onset of target motion. The three visual motion signals are summed and integrated to produce the eye velocity output of the model. The model reproduces the average eye velocity evoked by steps of target velocity in monkeys and humans and accounts for the variation among individual responses and subjects. When its motor pathways are expanded to include positive feedback of eye velocity and a “switch”, the model reproduces the exponential decay in eye velocity observed when a moving target stops. Manipulation of this expanded model can mimic the effects of stimulation and lesions in the arcuate pursuit area, the middle temporal visual area (MT), and the medial superior temporal visual area (MST).

## Introduction

Computer models of pursuit eye movements provide a powerful tool for investigation of the neural basis for pursuit. In the past, the development and simulation of quantitative models has identified data that place strong constraints on the organization of the neural pursuit system and has highlighted unresolved issues concerning the neural substrate for pursuit. Over the past several years, however, the description of novel properties in the pursuit system has largely outpaced the development of quantitative models. Recent experiments in humans and monkeys have provided a wealth of data about the dynamics of pursuit eye movements, the effects of lesions in various brain structures, and the physiological signals that are recorded in those structures.

Available data suggest three major criteria that should be satisfied by models of the pursuit system. First, the model should be able to replicate the dynamics of the pursuit eye velocity produced by the ramp target motions now used commonly to assess smooth pursuit behavior. Second, because the organization of the pursuit system is believed to be largely conserved across primates, a single class of model should be able to account for the pursuit behavior of both monkeys and

humans. Third, the model should be able to reproduce both the symptoms that follow lesions of different brain structures and the effects of either behavioral manipulations or electrical stimulation. Finally, the model should reflect the anatomy and neurophysiology of the pursuit system.

Recent models of pursuit have focused on the first criterion and have suggested ways to reproduce the trajectory of the eye velocity recorded in humans or monkeys during pursuit. In each model, the input to the system is target velocity and the output is eye velocity. Because the retina is attached to the moveable eyeball, the models are configured as negative feedback control systems that are driven by image motion, defined as target motion with respect to eye motion. The principle challenge addressed by the models was to reproduce both the rate of the initial rise in eye velocity at the initiation of pursuit and the correct frequency of oscillations in eye velocity during sustained pursuit of target motion at constant speed. One class of model, developed by Robinson et al. (1986) to fit data from human subjects, accomplishes this goal by using two separate mechanisms to control the initiation and maintenance phases of pursuit. The trajectory of eye velocity at the initiation of pursuit is

driven by visual inputs, whereas the oscillations during prolonged pursuit of target motion at constant speed are controlled by an internal feedback loop. Because the internal feedback loop determines the dynamics of the sustained eye velocity produced by this class of model, we refer to them as “internal feedback” models. A second class of model accomplishes similar goals by using visual input signals that are related to image acceleration as well as image velocity (Morris and Lisberger, 1985; Lisberger et al., 1987; Krauzlis and Lisberger, 1989). Because the dynamics of the eye velocity emitted by this class of model are determined by visual inputs, we refer to them as “image motion” models.

Recent experiments in monkeys have provided data that discriminate between existing “internal feedback” and the “image motion” models. In monkeys, increasing the time delay of visual feedback caused decreases in the frequency of oscillations in eye velocity during sustained pursuit of ramp target motion (Goldreich et al., 1992). An earlier version of our “image motion” model (Krauzlis and Lisberger, 1989) could reproduce the observed changes in the frequency of oscillation, while the “internal feedback” model of Robinson et al. (1986) could not. These results implied that image motion models based upon multiple visual inputs (Morris and Lisberger, 1985; Krauzlis and Lisberger, 1989) describe the operation of the pursuit system more accurately than do models based on internal feedback.

There are other features of normal pursuit that are not reproduced by either class of model. For example, most models of pursuit predict that eye velocity at the offset of pursuit should have the same profile as it does at the onset of pursuit. In fact, when a moving target stops, eye velocity usually returns smoothly to zero and does not display the overshoot and oscillations that occur in the initiation of pursuit (Robinson et al., 1986; Luebke and Robinson, 1988; Huebner et al., 1992; Krauzlis and Lisberger, 1994). These and other observations have led to the suggestion (Grasse and Lisberger, 1992; Goldreich et al., 1992) that a “pursuit switch” controls the transition between fixation and pursuit tracking. One manifestation of the switch is that the pursuit system can respond to sinusoidal target motion at frequencies above 2 Hz, but only if the target motions are presented after pursuit has been initiated (Goldreich et al., 1992; Schwartz and Lisberger, 1994). A second manifestation is that spontaneous oscillations in eye velocity occur frequently during pursuit of a moving target, but only rarely during fixation of a stationary target (Robinson, 1965).

In this paper, we present our “image motion” model of pursuit. Our goal was to describe a model that satisfied the first three of the four criteria set out above for models of pursuit. Thus, we have not attempted to develop a model of visual motion processing with biologically-realistic internal signals or anatomically-accurate connections. Rather, we describe a model that simulates the dynamics of the visual inputs for pursuit in some quantitative detail and in an easily adjustable structure. We demonstrate that the image motion class of model can account quantitatively on a time scale of milliseconds for the pursuit eye velocity of monkeys and humans. We also show how changes in the architecture of its motor pathways allow the model to reproduce eye velocity at the offset of pursuit as well as the results of stimulating or lesioning different brain structures that participate in pursuit.

## Methods

### *Computer Simulations*

The model is shown in Fig. 1. Its input was target velocity ( $\dot{T}$ ) and its output was eye velocity ( $\dot{E}$ ). We kept the model in the velocity domain partly because our data base consisted of measurements of eye velocity and partly so that we could focus on the transformation of visual inputs into commands for eye motion. We wished to avoid adding complexities that were irrelevant to this transformation, such as the neural integration that converts commands for eye velocity into the eye position signal needed by motoneurons, the forces developed in agonist and antagonist muscles, or the physical properties of the eyeball that determine the relation between muscle force and eye motion. In models that consider the neural integrator and the mechanics of the eyeball and muscles, the brain provides the inverse transfer function of the mechanical periphery. Therefore, the performance of our model was not affected by the omission of these elements. Previous models that operated in the velocity domain implemented the mechanical periphery as a filter with a single time constant of 15 ms (e.g., Robinson et al., 1986). In control simulations, we showed that the presence of this short time constant did not have a noticeable effect on the output from our model, because the filters within the model had comparable or longer time constants. Because its output was eye velocity, the model did not make saccades and could not change the position of the target with respect to the fovea. As a

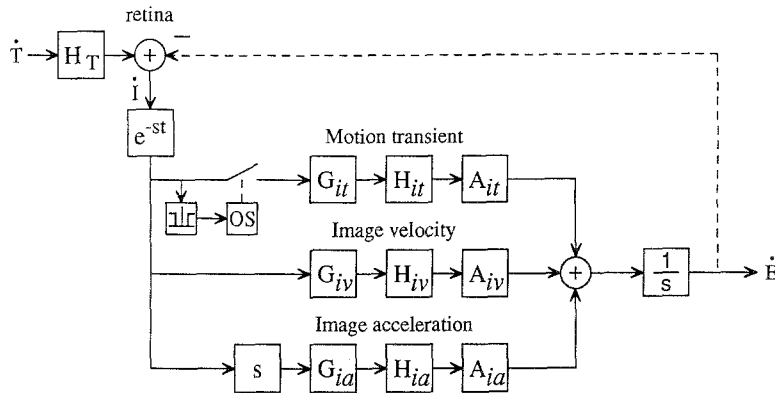


Fig. 1. Schematic diagram of the "image motion" model of pursuit. The inputs to the model are related to a motion onset transient, image velocity, and image acceleration.  $H_T$  is a transfer function that filters the target velocity input. The summing junction labeled "retina" computes image velocity ( $\dot{I}$ ) as target velocity ( $\dot{T}$ ) minus eye velocity ( $\dot{E}$ ). Image velocity is then time delayed by the function " $e^{-st}$ ". The box labeled "s" differentiates its inputs while that labeled " $1/s$ " performs mathematical integration.  $G_{it}$ ,  $G_{iv}$ , and  $G_{ia}$  are non-linear functions that serve as gain elements for the motion transient, image velocity, and image acceleration pathways, respectively.  $H_{it}$ ,  $H_{iv}$ , and  $H_{ia}$  are second-order filters that transform the signals in the three visual pathways.  $A_{it}$ ,  $A_{iv}$ , and  $A_{ia}$  are gain elements that allowed us to scale individually the outputs from the three pathways. The box labeled "OS" receives inputs from a thresholding non-linearity and emits a 20-ms duration pulse that controls a switch in the inputs to the motion transient pathway. The dashed line indicates external feedback of eye velocity, resulting from the fact that the retina is attached to the moving eyeball.

result, the model does not reproduce the observation that the magnitude of eye acceleration depends on the retinal position of the moving images (Lisberger and Westbrook, 1985; Tychsen and Lisberger, 1986). This omission should not have a large effect on the operation of the model because the database was obtained for images that were moving within the central  $4^\circ$  of the visual field, where the magnitude of eye acceleration depends only weakly on the position of the moving images (Lisberger and Westbrook, 1985). In addition, our data base included only target motion toward the position of fixation and we did not attempt to model the difference in the eye acceleration produced by target motion toward vs. away from the position of fixation (Lisberger and Westbrook, 1985).

Target velocity was first transformed by the function  $H_T$  before being passed on to the three visual pathways.  $H_T$  had the form:

$$H_T(s) = (\omega_T^2)/(s^2 + 2\xi_T\omega_Ts + \omega_T^2)$$

where  $\omega_T$  describes the cutoff frequency in radians and  $\xi_T$  gives the damping ratio. The function of  $H_T$  was to low-pass filter the idealized steps in target velocity provided as inputs. The filtering allowed us to match the input to the visual pathways of the models more closely to the actual dynamics of the target motions provided by the mirror-galvanometers used in our experiments. For all of the simulations we set the values

of  $\omega_T$  and  $\xi_T$  equal to 200 and 0.9, respectively. These parameters converted a step input into a smooth output that rose to  $1/e$  of its final value in 10 ms with no overshoot. The visual input to the model was obtained by subtracting  $\dot{E}$  from the output of  $H_T$  at the summing junction labeled "retina" to produce a signal related to image velocity ( $\dot{I}$ ). This subtraction gives the model the basic negative-feedback organization of the pursuit system. The output from the "retina" summing junction was then passed through a time delay ( $e^{-st}$ ). We did not attempt to model the factors responsible for the relationships between latency and image speed or acceleration. Instead, we aligned all of our data traces on a single latency of 80 ms and set the delay ( $t$ ) to a value of 60 ms. The filtering properties of other elements of the model contributed the additional 20 ms of delay needed to give the output of the model a total latency of 80 ms.

The model had 3 parallel visual pathways that provided outputs related to image velocity, image acceleration, and motion onset transient (Krauzlis and Lisberger, 1989, 1994). Each pathway consisted of a non-linearity that modified the amplitude of each signal, a filter that adjusted the time courses of the signals, and a gain element. The gain elements, designated as  $A_{it}$ ,  $A_{iv}$ , and  $A_{ia}$ , were used to adjust the overall strength of each pathway. The three non-linearities, designated  $G_{it}$ ,  $G_{iv}$ , and  $G_{ia}$ , were implemented by

applying a cubic spline interpolation algorithm (Press et al., 1988) to three sets of data points relating eye acceleration to the three parameters of image motion. Each non-linearity was initialized with values of eye acceleration obtained in our experiments on monkeys (Krauzlis and Lisberger, 1994). Although these data provided a good starting point for the model, changes in the non-linearities were required to obtain optimal fits of the model to the eye velocity of monkeys. We customized the non-linearities by changing the values of eye acceleration in a given set of points and computing a new cubic spline interpolation that then was used to run the model. Two precautions ensured that this approach did not devolve into a strategy similar to the use of a lookup table. First, we restricted the number of data points used for the interpolation to fourteen for the image velocity and image motion transient pathways, and to sixteen for the image acceleration pathway. For the image velocity and image motion transient pathways, these data points were located at  $\pm 2.5, 5, 10, 15, 20, 25$ , and  $50^\circ/\text{s}$ . For the image acceleration pathway, these data points were located at  $\pm 25, 45, 64, 80, 180, 320, 400$ , and  $1000^\circ/\text{s}^2$ . Second, we restricted the number of inflections included in the functions described by the data points. In most cases, functions contained no more than four points of inflection. Four inflections allowed a function to have a slope equal to zero as it passed through the origin and to be tuned for both positive and negative values of the input variable. These functions were designed to allow the model to employ highly-flexible non-linearities while ensuring reasonable generalization as the model interpolated between the points used to define each cubic spline. For the purpose of this paper, which was to demonstrate the plausibility of a model based on three different visual motion signals, we did not attempt to match the non-linearities in the model to those in the responses of neurons in the cortical motion pathways. Future efforts will have to take the next step of ensuring that the visual signals recorded in the model resemble those recorded in the brain.

The transfer functions used as filters in the visual pathways, designated  $H_{iv}$ ,  $H_{ia}$ , and  $H_{it}$ , each had the form:

$$H(s) = (\omega^2)/(s^2 + 2\xi\omega s + \omega^2)$$

where  $\omega$  describes the cutoff frequency in radians/s and  $\xi$  gives the damping ratio. We initially used first-order systems as the filters in each of the pathways, but we

found that the ability to transform visual inputs with an underdamped filter assisted the model in reproducing some of the overshoot observed in the eye velocity responses of monkeys. We do not wish to imply that such filters are implemented explicitly in the pathways that guide pursuit, but we think it is plausible that the process of extracting information about image direction, speed, and acceleration could be represented by such filtering.

The image acceleration and the motion onset transient pathways both contained elements in addition to the gain element, non-linearity, and transfer function. For the image acceleration pathway, the differentiator before the non-linearity converted the image velocity input into a signal related to image acceleration. It was possible to use a differentiator at the front of the image acceleration pathway without rendering the performance of the model sensitive to the size of the simulation time step because  $H_T$  converted the step inputs to the model into smooth changes in target velocity. For the motion transient pathway, a non-retriggerable one-shot ("OS" in Fig. 1) and a threshold element preceded the non-linearity. When image velocity was zero, the output of the threshold element was zero and the motion transient pathway remained quiescent. When the absolute value of image velocity exceeded a threshold of  $3.5^\circ/\text{s}$ , the one-shot closed a switch for 20 ms to allow a pulse derived from the image velocity signal to pass to the gain element in the pathway. After the fixed time had elapsed, the switch opened and no further inputs were allowed to pass through the motion onset transient pathway for the remainder of the target motion. This combination of elements was designed to provide a brief pulse of image velocity at the onset of a step in target velocity. Future research will have to determine how such a signal is represented in the brain and how that representation is formed in the pathways that mediate pursuit.

To generate the eye velocity output from the model, the outputs of the visual pathways were summed and integrated. In the initial version of the model, the integration was performed by a perfect integrator, as shown in Fig. 1. In a later version of the model, which will be described in the second half of the Results, the perfect integrator was replaced with a "leaky" integrator. The leaky integrator consisted of an exponential decay with a single time constant (a single "pole") and a gain element.

Individual parameters in the model were adjusted with one of two methods that we refer to as "adjusted by hand" and "optimized". When parameters were

“adjusted by hand”, the user assigned values directly to individual parameters. When parameters were “optimized”, the user marked some or all of the parameters and specified a set of eye velocity data to which the output of the model was to be matched. The marked parameter values were then adjusted iteratively using the downhill simplex method of Nelder and Mead (1965) to minimize the value returned by an error function defined as the sum over the interval in question of the absolute values of the differences between the output of the model and the data from the monkey. After optimization was complete, we used several methods to characterize the performance of the models. First, we quantified how well each model fitted the experimental data by measuring the “average absolute error”, defined as the sum of the absolute values of the differences between actual and simulated eye velocity at each millisecond in a selected interval divided by the duration of the interval. We used this measurement, rather than mean squared error, because it retains the units of the original data ( $^{\circ}/s$ ). Second, we added additional inputs to individual nodes in the model to test the effects of stimulation at particular sites. Finally, we plotted the time course of the eye velocity output from the model.

#### *Database for the Model*

The parameters of the model were adjusted so that its output matched data reported in Krauzlis and Lisberger (1994). Briefly, data were obtained from three rhesus monkeys that were rewarded for fixating and tracking small targets that moved across a tangent screen. Methods that have been described elsewhere were used to present target motion in trials and to record eye position, target position, and eye velocity (Lisberger and Westbrook, 1985; Krauzlis and Lisberger, 1994). Each trial began when the monkey fixated a red target that was stationary at straight-ahead gaze. After the monkey had fixated for 300 ms, a white target appeared at an eccentric position. The monkey was required to continue fixating the red target for a randomized interval of 300 to 600 ms. Then, the red target was extinguished and the white target began to move. In some trials, the target underwent a step change in velocity up to speeds of 5, 10, 15, 20, or 25 $^{\circ}/s$ . In other trials, the target accelerated at constant rates for a period of 125 ms up to the same sets of speeds. For experimental trials, the target moved from eccentric positions toward the position of fixation. For control trials, target motion was away from the position of fixation. Different target motions

and control trials were presented in a random order so that the monkeys could not anticipate correctly the direction, trajectory, or final speed of target motion. For the computer simulations in this paper, we attempted to match the output of the model only to the eye velocity in trials that presented steps of target velocity.

Before analyzing the data, the records for each trial were displayed on a video screen and the eye velocity traces were edited under the control of the experimenter. A tiny proportion of the trials were excluded from the data analysis if the monkey made a saccade within 200 ms after the onset of target motion. We excised the rapid deflections of eye velocity associated with other saccades and replaced them with straight-line interpolations between smooth eye velocity at the start and end of the saccade. Saccades were small in amplitude and therefore short in duration, so less than 50 ms of data was replaced by interpolation. We think this procedure is valid because of the similarity of the trajectory of eye velocity in trials that did and did not contain small saccades. Trials were then sorted according to the final speed and direction of target motion and the records from identical trials were aligned on the onset of target motion and averaged.

#### **Results**

We report on three steps in the development of our image motion model. First, we document parameters of the model that matched its output to the first 380 ms of eye velocity evoked by steps in target velocity from zero to 5, 10, 15, 20, and 25 $^{\circ}/s$  in three monkeys. Because the eye velocity of the different monkeys followed quite different trajectories for the same stimulus, we allowed the model to have different parameters for different individuals, but we maintained the same structure of the model for all individuals. Next, we demonstrate that the parameters of our model could be adjusted to reproduce the performance of a different class of model devised by Robinson et al. (1986) to account for the eye movements evoked by steps of target velocity in humans. Finally, we elaborate the structure of the model so that its output could also reproduce the eye velocity evoked at the offset of ramp target motion.

#### *Modeling the Average Eye Velocity of Three Monkeys*

The good agreement between the outputs of the models (dashed traces) and the eye velocity averages from

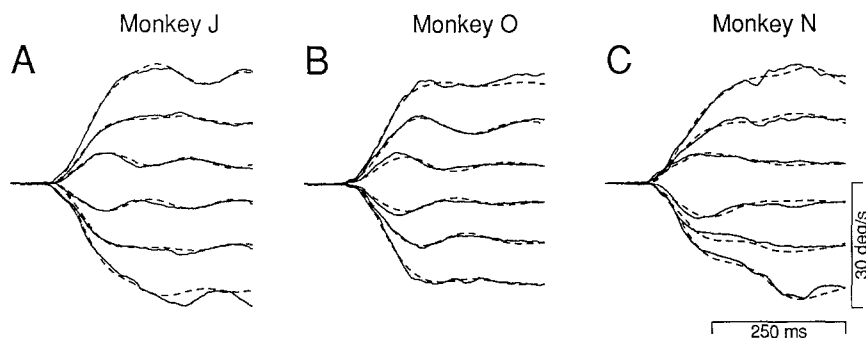


Fig. 2. Performance of the optimized models for all three monkeys. The dashed traces show the output from the model after it had been optimized to fit the first 380 ms of pursuit. The solid traces show the averaged eye velocity of the three monkeys for rightward and leftward steps of target velocity from zero to 5, 15, and 25°/s. Upward deflections of the traces indicate rightward eye motion.

three monkeys (solid traces) is shown in Fig. 2. There are three types of features that the model was able to reproduce well. First, the model matched the initial rise in eye velocity at the onset of pursuit. The use of different parameters in the non-linearities for the visual pathways allowed the model to reproduce the initial eye velocities of monkeys with high (monkey O), medium (monkey J), and low (monkey N) eye accelerations during the initiation of pursuit. Second, the model reproduced the transient overshoot in eye velocity during the transition to sustained pursuit. The use of different parameters in the filters for the visual pathways was able to reproduce large (monkeys J and O) and smaller (monkey N) overshoots in eye velocity during the transition to sustained tracking. Third, the model reproduced reasonably well the oscillations that occurred during sustained pursuit in the three monkeys.

There were also several details of the data that the model did not capture very well. Most of these were instances where the features of the eye velocity were not uniform for both rightward and leftward pursuit, shown by upward and downward deflections of the traces. In Fig. 2A, for example, the model did not reproduce well the oscillations in eye velocity for target motion to the left at 25°/s, even though it did well for target motion to the right at the same speed. In Fig. 2C, the model did not match the different dynamics for rightward versus leftward pursuit in monkey N. The monkey's pursuit of a target moving to the right at 25°/s showed a smooth continuous acceleration, while target motion to the left at 25°/s elicited two distinct phases of eye acceleration. In addition, the transition from initiation to steady-state pursuit showed a more pronounced overshoot for leftward than for rightward target motion at 5°/s in this monkey. The model was still able to reproduce the

general features of the eye velocity in monkey N, but did not match both rightward and leftward trajectories with as great accuracy as we would have liked. In a small series of test simulations, we found that better fits were possible if separate sets of filter parameters were used for rightward and leftward pursuit. However, we did not think that the small improvements in performance were worth the increase in the complexity of the model.

Figure 3 shows the cubic spline functions that served as non-linearities in the three visual pathways for the models that provided the best fits to the data for the three monkeys. In each monkey, the function in the image velocity pathway (Figs. 3A, D, G) was close to linear over the range of target velocities used in our experiments (−25 to 25°/s). The functions in the motion onset transient pathway (Figs. 3B, E, H) were consistently non-linear. They generally contained a region with a steeper slope for low velocities of image motion and started to saturate for higher velocities of image motion. The functions in the image acceleration pathways (Figs. 3C, F, I) were similar in all three monkeys. Each showed a region of high slope for small image accelerations, reached positive or negative peaks for intermediate image accelerations, and returned almost to zero for the highest image accelerations. A narrow region of very low sensitivity was a consistent feature of almost all of the non-linearities in the image velocity and image acceleration pathways (Figs. 3A, C, D, F, G, I). This small “dead-zone” proved to be an important feature in allowing the models to reproduce accurately the averaged eye velocity recorded in our monkeys during pursuit of sustained target motion. Table 1 reports the parameters used in the filters for the models used to generate Figs. 2 and 3.

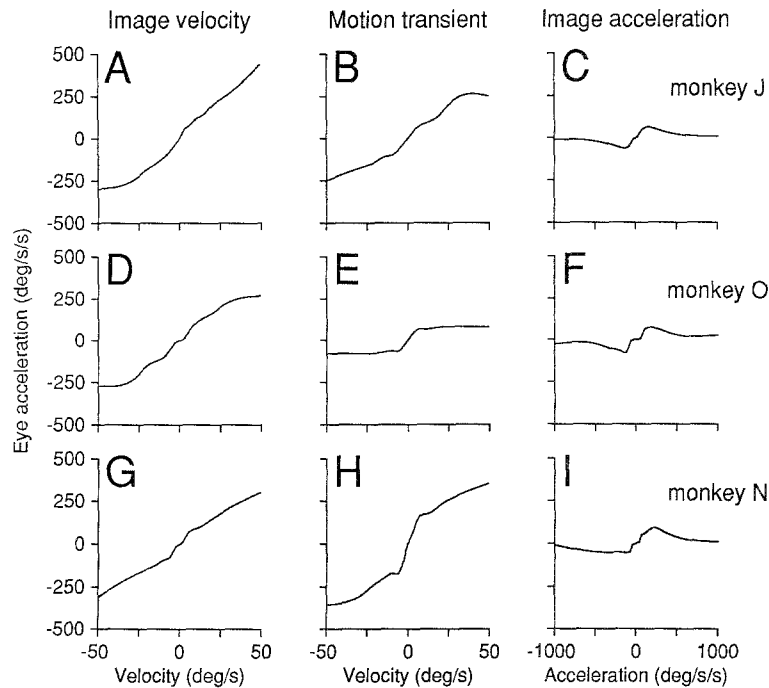


Fig. 3. Non-linear functions used in the optimized models. Each graph shows the cubic spline function that was used to transform the image motion inputs into commands for smooth eye acceleration. The three rows of graphs show the functions used to fit the data of monkey J (A, B, C), monkey O (D, E, F), and monkey N (G, H, I). The three columns of graphs show the functions used for the image velocity pathway (A, D, G), the motion onset transient pathway (B, E, H), and the image acceleration pathway (C, F, I).

Table 1. Parameters of the filters in the visual motion pathways for the optimized three-pathway models. The outputs of the models are shown in Fig. 2 for the three monkeys and in Fig. 8 for the simulations of pursuit in humans. The equation that describes the filters appears in the Methods.

|        | Image velocity |       | Image acceleration |       | Motion transient |       |
|--------|----------------|-------|--------------------|-------|------------------|-------|
|        | $\omega$       | $\xi$ | $\omega$           | $\xi$ | $\omega$         | $\xi$ |
| Monkey |                |       |                    |       |                  |       |
| O      | 34.5           | 0.49  | 35.3               | 0.24  | 38.8             | 0.20  |
| J      | 50.4           | 0.76  | 50.7               | 0.14  | 28.7             | 0.35  |
| N      | 22.0           | 0.37  | 69.0               | 0.20  | 28.1             | 0.11  |
| Human  | 16.2           | 0.51  | 20.7               | 0.21  | 27.8             | 0.45  |

#### Modeling the Trial-to-Trial Variability in the Trajectory of Eye Velocity

Changes in the gains of the three visual motion pathways in the model were able to account for much of the trial-to-trial variation in the trajectory of eye velocity in individual responses to the same target motion in a given monkey. The solid trace labeled with an asterisk in Fig. 4A shows the averaged eye velocity for fourteen presentations of a step of target velocity from

zero to 10°/s rightward on one day in monkey J. After the model for monkey J had been optimized to fit the eye velocity evoked by the full set of steps of target velocities, its output for a 10°/s target motion (dashed trace) followed the averaged data closely. To fit the eye velocity of each of the 14 individual trials used to form the average, we ran the optimization procedure again for each individual trial, but we allowed it to adjust only the values of three scaling factors placed at the outputs of each of the three visual motion pathways. Allowing the model to adjust only the scaling factors placed some constraints on how it reproduced the data for individual trials. If we had allowed different values of all the parameters for each individual trial, then the model would have done a better job of reproducing the data from our monkeys.

The traces labeled "X", "Y", and "Z" in Fig. 4A show that optimization of the scaling factors was able to produce an output from the model (dashed lines) that accounted for the varying amounts of ringing and overshoot in the eye velocity traces recorded in the monkey (solid lines). Figures 4B and C report the values of the scaling factors used in the three pathways of the model

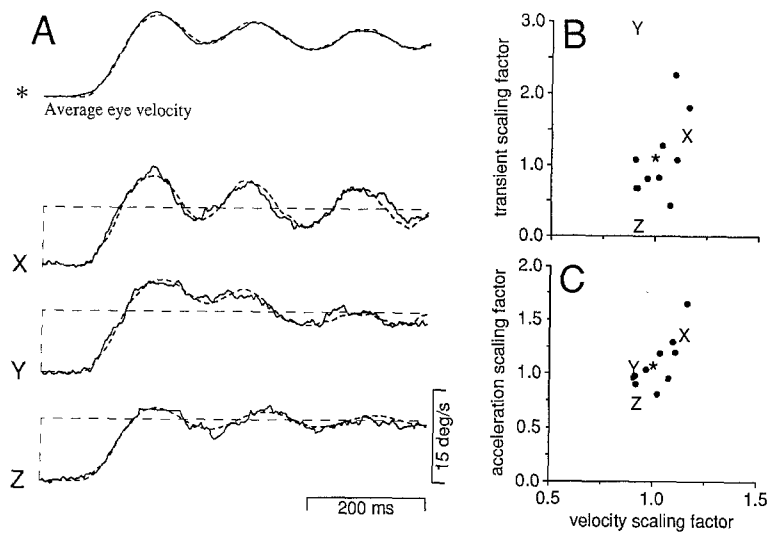


Fig. 4. Changes in the scaling of the signals from the three visual motion pathways model the trial-to-trial variability in the trajectory of eye velocity. A. The dashed traces show the performance of the model and the solid traces show the eye velocity from monkey J. The trace labeled with an asterisk shows averaged eye velocity for 14 responses to a step of target velocity from zero to  $10^\circ/\text{s}$  rightward. The traces labeled "X", "Y", and "Z" show the eye velocity in 3 of the 14 individual trials that were used to form the average. The model has been optimized to fit each individual trial by adjusting the relative scaling of the signals from the three visual motion pathways. B, C: Each point plots the scaling factors used to fit the model to each of the 14 individual trials. Each graph shows the scaling factor for the motion onset transient pathway (B) and the image acceleration pathway (C) as a function of that for the image velocity pathway. The asterisk indicates that the scaling factors ( $A_{iv}$ ,  $A_{il}$ ,  $A_{ia}$ ) were 1.0 in all three pathways when the model was optimized to fit average eye velocity. The symbols X, Y, and Z in the graphs show the scaling factors that provided the best fit to the three eye velocity individual traces in A.

to match the eye velocity from all fourteen trials. The image velocity scaling factor required the least amount of change to fit the responses to the 14 individual trials, while the scale factors for image acceleration and motion onset transient varied quite widely. The points marked by letters, which correspond to the three eye velocity traces shown in Fig. 4A, illustrate how scaling of the outputs from the different visual pathways in the model was associated with particular features of eye velocity. For example, the two traces with the largest overshoot in the initial eye velocity (X and Y) had the largest scaling factors for the image motion transient (Fig. 4B). Trace X showed large oscillations that continued through the course of the trial and was modeled with a large scaling factor for image acceleration (Fig. 4C). Trace Z had low scaling factors for both the image acceleration pathway and the motion onset transient pathway and showed only a small amount of overshoot and the smallest oscillations in sustained eye velocity. Thus, high gain in the motion onset transient pathway causes overshoot in the initiation of pursuit while high gain in the image acceleration pathway promotes large, persistent oscillations of eye velocity during sustained pursuit.

We next assessed the time course of the changes in the scaling factors required to fit the eye velocity in individual trials over the 1 to 2 hour course of a single experiment. Once again, we first optimized the parameters in the model to match the averaged eye velocity recorded during the first 400 ms of pursuit for steps of target velocity from zero to 5, 10, 15, 20 and  $25^\circ/\text{s}$  rightward and leftward. The individual eye velocity records used to calculate these averages were obtained in 400 consecutive trials from an experiment that consisted of slightly over 1300 trials and lasted about 2 hours. The randomized sequence of trials included steps of target velocity to the right and left and other target motions such as smooth target accelerations (see Krauzlis and Lisberger 1994). For each of the individual trials that presented steps of target velocity ( $n = 99$  of 400 total trials), we optimized the values of the scaling factors in the three visual pathways so that the output from the model matched the eye velocity recorded from the monkey. Figure 5D shows that this procedure produced relatively good fits: the averaged absolute error was less than  $1^\circ/\text{s}$  in over 50% of the trials and averaged  $1.075 (\pm 0.54 \text{ standard deviation})$ . Figures 5A–C plot the values of the scaling factors obtained for each



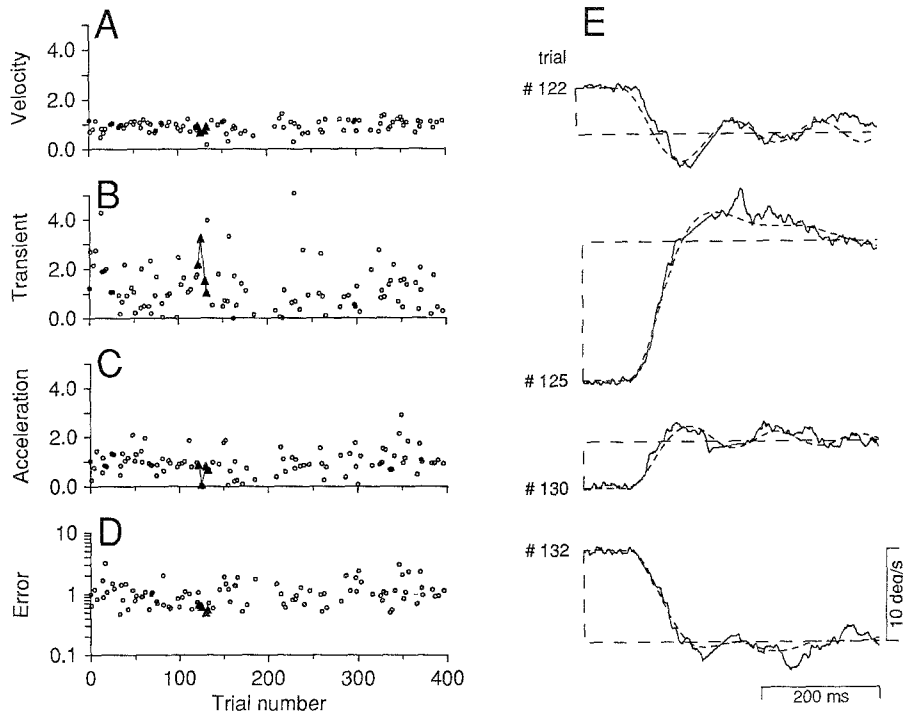


Fig. 5. Time course of variation in the scaling factors used to fit the model to each individual trial in one daily experiment. A–C: Each point plots the value of the scaling factor in the image motion transient (A), image acceleration (B), or image velocity pathway (C) as a function of the number of the trial during the experiment. D: Average absolute error of the fit to each individual trial is plotted as a function of the number of the trial during the experiment. E: Eye velocity evoked in the four consecutive trials indicated by the connected filled triangles in A–D. Solid traces show the eye velocity recorded in the monkey and dashed traces show the output of the model after the scaling factors had been adjusted to optimize the performance on each trial. The target velocity in these four trials was 5°/s leftward (#122), 15°/s rightward (#125), 5°/s rightward (#130), and 10°/s leftward (#132). Data are from monkey J.

of the three pathways as a function of trial number for the 99 steps of target velocity included in this experiment. The values of the scaling factors varied widely among trials, again with the largest variation in the image acceleration and image motion transient pathways (Figs. 5B and C) and the least variation in the image velocity pathway (Fig. 5A). To provide an example of this trial-to-trial variation, the traces in Fig. 5E show data and model outputs for the four successive steps of target velocity. The scaling factors required to match the output of the model (dashed lines) to the monkey's eye velocity (solid lines) for these sample trials are indicated in Figs. 5A–C by the four triangles that have been connected by lines. This analysis revealed that both the scaling factors and the eye velocity records could show large changes over very few trials. In the sequence of trials illustrated in Fig. 5E, a trial with pronounced overshoot and large oscillations during sustained pursuit (#122, 5°/s leftward) was followed in rapid succession by a trial with a large sustained over-

shoot (trial #125, 15°/s rightward) and two trials (#130 and #132) with much less overshoot and oscillation. In Fig. 5E, the averaged absolute error was 0.69, 0.63, 0.50, and 0.56 for trials #122, #125, #130, and #132, respectively.

For monkey O, we were able to follow the eye velocity records evoked by the same target motions over the course of two months. We recorded his eye movements on the first day he tracked steps of target velocity ("day 0"), as well as on the next day and on several subsequent days. We analyzed these data by first optimizing the parameters in the model to match the averaged eye velocity records obtained on an experiment performed 22 days after the first experiment. We then held the parameters in the non-linear functions and filters fixed and optimized the scaling factors at the output of the three visual pathways to match the data obtained on days preceding and following day #22. For each day, we adjusted the values of the scaling factors so that the model reproduced averages of eye velocity

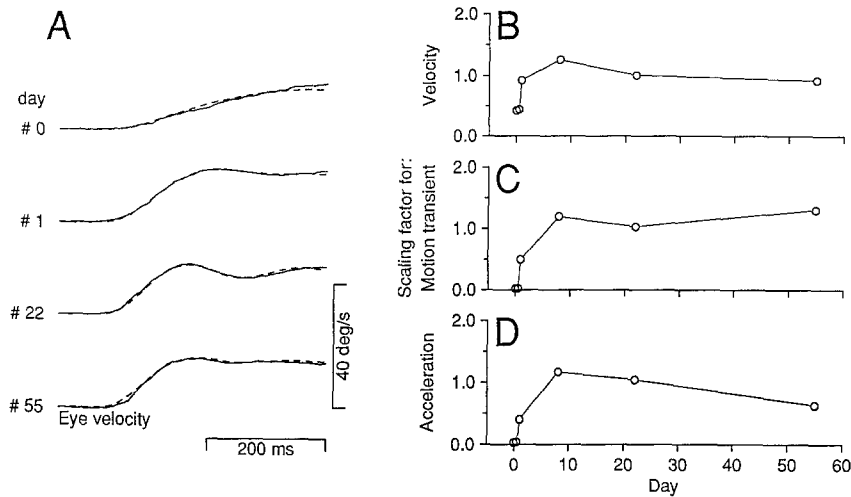


Fig. 6. Systematic changes in the scaling of the three visual motion pathways over a time course of 55 days. A: Responses to a step of target velocity from zero to 15°/s rightward on four days of experiments in monkey O. The solid lines show averaged eye velocity and the dashed lines show the output from the model after it had been optimized by adjusting the scaling factors in the three visual pathways. The numbers to the left of each trace indicate the experimental day relative to the first day steps of target velocity were shown to the monkey. The traces shown for day #0 represent the averaged data obtained during the first half of the experimental session. B–D: Each point shows the scaling factors used to optimize the performance of the model for averages of eye velocity from different days. The graphs plot the scaling factors in the image velocity pathway (B), motion onset transient pathway (C), and image acceleration pathway (D) as a function of the number of days since the start of training. The two points for day #0 show the results of analyzing the first and second halves of the data separately.

recorded during pursuit of targets moving at constant velocities of 5, 10, 15, 20 and 25°/s. For day #0, we generated separate averages and obtained separate values of the scaling factors for the data obtained during the first and second halves of the experiment.

Figure 6A shows the good agreement between the monkey's average eye velocity on different days (solid lines) and the performance of the model (dashed lines) for target motion at 15°/s. Figures 6B–D plot the values of the scaling factors obtained for each of the three pathways as a function of experimental day. The latency of pursuit was nearly the same on days 0 through 55. However, the trajectory of eye velocity and the values of the scaling factors showed consistent changes as a function of time. On day #0, the initial rise in eye velocity was slow and could be modeled using just the image velocity pathway with a low overall gain; the optimization algorithm zeroed the scaling factors in the image acceleration and image motion transient pathways. The two superimposed data points in the graphs for day #0 demonstrate there were no differences in the results obtained with data from the first and second halves of the experiment. On day #1, the faster rise in the monkey's eye velocity could be fitted by assigning a larger gain to the image velocity pathway and by

partial use of the image acceleration and motion onset transient pathways. By day #8, all three scaling factors and the trajectory of eye velocity had reached values typical of the performance of this monkey throughout the remainder of the experiments. After day #8, neither the scaling factors used in the model nor the trajectory of the averaged eye velocity records showed any further major changes. Unfortunately, we did not have data that would have allowed us to evaluate whether the decreased scaling factor for image acceleration on day #55 represented a trend.

#### *Simulation of Oscillations in Eye Velocity During Maintained Pursuit in Monkeys*

Experiments on monkeys have shown that eye velocity oscillates around target velocity during sustained pursuit of target motion at constant speed and that the period of the oscillation increases when the time delay in the visual input for pursuit is increased (Goldreich et al., 1992). Figure 7 demonstrates that increasing the visual delay has the same effect on the performance of our model. The bottom trace in Fig. 7A shows the output from the model when the visual delay was

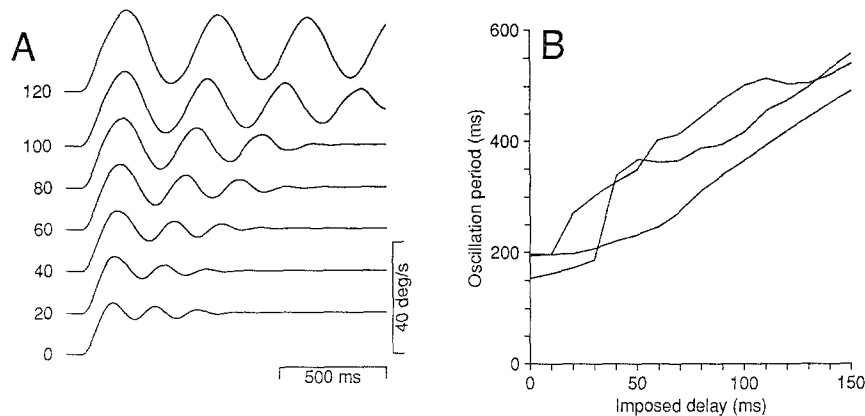


Fig. 7. Effect of increasing the visual feedback delay on oscillations in pursuit eye velocity. A: Each trace shows the output from the model when additional delay was imposed. The numbers next to each trace indicate the number of milliseconds of additional delay used to obtain that trace. B: Each curve plots the period of oscillation in the model as a function of the additional delay imposed in the feedback loop. The three curves were obtained using the models that had been optimized to reproduce the eye velocity of monkeys J, N, and O.

60 ms and the other parameters were the same as those used to generate Fig. 2B, except that the outputs of all three visual pathways were scaled by a factor of 1.2 to emphasize the oscillations in eye velocity. The other traces show the output of the model when the external feedback loop (dashed line in Fig. 1) included extra milliseconds of delay indicated by the numbers to the left of each trace in Fig. 7A. As the imposed delay was increased, both the period and the amplitude of the oscillations increased. Figure 7B plots the major oscillation period in the output of the model as a function of the delay that was imposed in eye velocity feedback for models fit to data from the three monkeys. We determined the major oscillation period by fitting a sine wave to the output from the model over the interval of sustained eye velocity. For each monkey, the output from the simulations showed a progressive increase in oscillation period as the imposed delay was increased. The general shape of the curves and the slope of the relationship between oscillation period and imposed delay agree well with the data from experiments on monkeys (Goldreich et al., 1992).

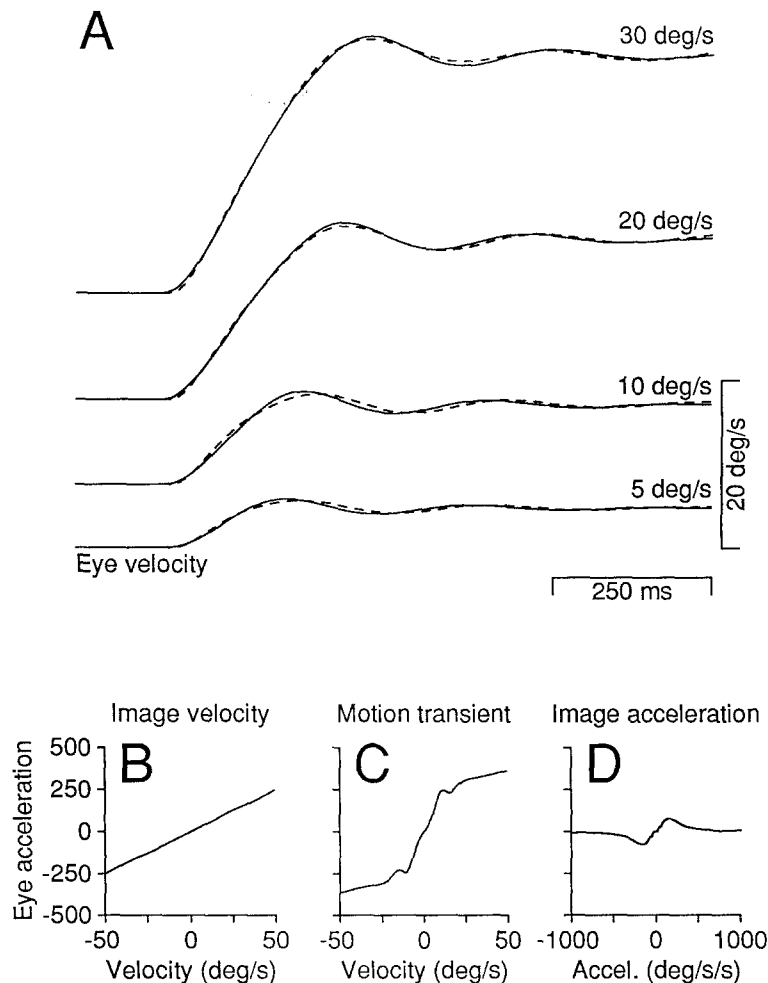
#### *Simulation of the Pursuit Performance of Humans*

Figure 8 shows that the class of model described here can account for the data on human pursuit that was the basis for the model of Robinson et al. (1986). Because the method used by Robinson et al. (1986) to analyze the trajectory of eye velocity eliminated the detailed time course of eye velocity required for our optimization procedure, we addressed this issue by adjusting the

parameters of our model to reproduce the output of the model of Robinson et al. (1986). The dashed traces in Fig. 8A show the output of their model for target motion at 5, 10, 20, and 30°/s, using the parameters given in their paper. The solid traces in Fig. 8A show that our model was able to achieve a nearly identical performance after the parameters had been optimized in the way described earlier for the data from monkeys. In particular, the output of the model matches the initial rising phase of the dashed traces and shows oscillations in simulated eye velocity during sustained pursuit at a frequency (3.6 Hz) that is close to that observed in human subjects. Figures 8B, C, and D show the non-linearities used in the image velocity, motion onset transient, and image acceleration pathways of the model that produced the simulations in Fig. 8A. The functions in all three pathways were similar to those in the models used to fit the data from our three monkeys (see Fig. 3). Table 1 reports the values of the filter parameters in the model that was used to obtain Fig. 8A.

#### *Simulation of the Offset of Pursuit*

It has been demonstrated previously in humans (Robinson et al., 1986; Luebke and Robinson, 1988; Huebner et al., 1992) and monkeys (Krauzlis and Lisberger, 1994) that the trajectory of eye velocity at the offset of pursuit is qualitatively different from that at the onset of the pursuit. While the onset of pursuit is non-linear and shows overshoot and oscillation, the eye velocity at the offset of pursuit scales



**Fig. 8.** Performance of our image motion model after its parameters had been optimized to reproduce the pursuit eye movements of human subjects. **A:** The solid lines show the output from the model published by Robinson et al. (1986) to account for the data of human subjects. The dashed traces show the output of the model presented in this paper after its parameters had been optimized to reproduce the output from the model of Robinson et al. (1986). Each pair of traces compares the responses of the two models for steps of target velocity. The number next to each pair of traces gives the final target velocity (5, 10, 20, and 30°/s). **B:** Non-linear function relating eye acceleration to image velocity for the image velocity pathway. **C:** Non-linear function relating eye acceleration to image velocity for the motion onset transient pathway. **D:** Non-linear function relating eye acceleration to image acceleration for the image acceleration pathway.

linearly with target velocity and shows an exponential decay to zero eye velocity. Figure 9A shows that the model we have analyzed so far, when optimized to reproduce eye velocity during the onset of pursuit and sustained tracking, cannot replicate the decay in eye velocity observed at the offset of pursuit. The performance of the model (dashed traces) fails to follow the eye velocity of the monkey (solid traces) accurately at the offset of pursuit (after the downward arrow in Fig. 9A) for any of the target velocities we used. In each example, the output of the model for

the offset of rightward target motion resembled its output for the onset of leftward target motion, and vice versa. Simple changes in the visual motion signals that contributed to the offset of pursuit did not cause the model to match the observed decay in eye velocity. To obtain the records in Fig' 9A, we zeroed the gain of the motion onset transient pathway after pursuit had been initiated, so that the motion onset transient did not contribute at the offset of pursuit. If the motion onset transient pathway had been used in the model when the target stopped, the overshoot of

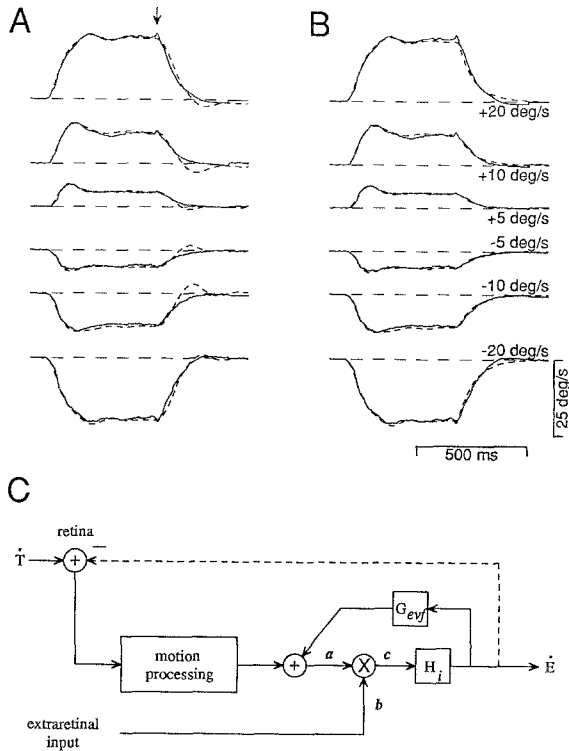


Fig. 9. A revised model that can account for the eye velocity at both the onset and the offset pursuit. A, B: The solid traces show the average eye velocity of monkey O for a step-ramp-step-stop target motion that started at zero velocity, moved at 5, 10, or 20°/s to the right or the left for 500 ms, and then stopped. In A, the dashed traces show the output of the model of Fig. 1 after it had been optimized to reproduce the first 380 ms of pursuit in monkey O. The downward arrow indicates the beginning of the offset of pursuit. In B, the dashed traces show the output of a revised model. C: A new version of the model that has been elaborated so it can simulate both the onset and the offset of pursuit. The visual inputs to the model and the three visual motion pathways are the same as those in Fig. 1 and are summarized in the box labeled “motion processing”.  $H_i$  is a leaky integrator described by a first-order system,  $H_i(s) = \tau/(s\tau + 1)$ .  $G_{evf}$  is a linear gain element that operates in positive feedback around  $H_i(s)$ . The  $a$ ,  $b$ , and  $c$  next to three of the pathways indicate the values of the signals carried along the inputs to and output from the multiplying junction labeled “X”, which computes  $c = a * b$ . The value of  $b$  was set to be 1 or 0 depending on whether the target was moving or stationary.

the model at the offset of pursuit would have been still larger and the failure of the model to reproduce the eye velocity from the monkey would have been more pronounced.

To simulate both the onset and offset of pursuit, we designed a model that switches between non-linear and linear modes depending on whether the tracking target is moving or stationary. Figure 9C shows a variation of

our model that retains all the features of Fig. 1 and adds two features to allow the model to reproduce the offset of pursuit. In Fig. 9C, a single box labeled “motion processing” represents three visual pathways; the detailed transfer functions for this part of the model were not modified from the foregoing analysis. In Fig. 9C, we implemented the mathematical integrator ( $1/s$ ) of Fig. 1 as a leaky integrator ( $H_i$ ) that was improved by positive feedback through gain element  $G_{evf}$ . In addition, we placed a multiplying junction in the input to  $H_i$  so that the output from the pursuit system (signal  $a$ ) was multiplied by the value of signal  $b$  before being forwarded to the leaky integrator. If we set the time constant of integration in  $H_i$  equal to 72 ms,  $G_{evf}$  equal to 13.9, and  $b$  equal to 1, then the model in Fig. 9C performed exactly like the model in Fig. 1 during the initiation of pursuit and sustained tracking of target motion at constant speed.

We simulated the onset and offset of pursuit by assuming that signal  $b$  acted as a switch. When the target was stationary, the value of  $b$  was zero. At a latency of 60 ms after the target started to move, the value of  $b$  was set to equal to one. At the same latency after the moving target was stopped, the value of  $b$  was set to zero. Figure 9B shows that the output of the model (dashed traces) was now able to reproduce the averaged eye velocity of the monkey (solid traces) for both the onset and offset of target motion at speeds of 5, 10, and 20°/s to the right and the left. This version of the model worked because changes in the value of  $b$  switched the model between one state that allowed non-linear visual inputs to control eye motion when the target was moving and a second state that placed eye velocity under the complete control of the linear leaky integrator ( $H_i$ ) when the target was stationary. In the model, the time constant of  $H_i$  (72 ms) determined the rate at which eye velocity decayed toward zero at the offset of pursuit.

The eye velocity positive feedback circuit and the multiplying junction in Fig. 9C were designed to reproduce the exponential decay to zero eye velocity observed at the offset of pursuit. We now show that this organization allows the model to mimic some of the effects of stimulating or ablating the pathways for pursuit eye movements in monkeys. For example, Fig. 10 shows the effects of stimulating the model in Fig. 9C at the sites labeled  $a$ ,  $b$ , or  $c$  during pursuit of target motion at 30°/s (Fig. 10A, B, C) or during fixation (Fig. 10D, E, F). During pursuit (Fig. 10A, B, C), increasing the activity of signal  $a$ ,  $b$ , or  $c$  causes a smooth eye velocity (solid trace) that clearly deviated from the

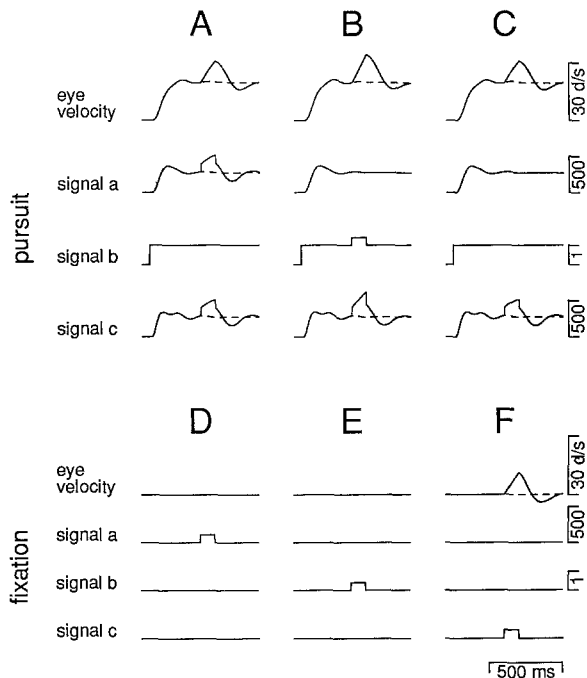


Fig. 10. Effect of stimulating signals a, b, and c on the output from the model. A–C: Stimulation was applied during target motion at  $20^\circ/\text{s}$  (“pursuit”). D–F: Stimulation was applied during fixation of a stationary target (“fixation”). The dashed lines indicate the output of the model for the same target motion in the absence of any stimulation. The four traces in each panel show the simulated eye velocity produced by the model and the activity present at the three locations in the model indicated by the letter labels a, b, and c in Fig. 9C. In A and D, stimulation caused a transient elevation of signal a. In B and E, stimulation caused a transient elevation of signal b. In C and F, stimulation caused a transient elevation of signal c.

control output of the model without stimulation (dashed traces). During fixation, in contrast, increasing the activity of signal **a** or **b** has no effect on eye velocity (Fig. 10D and E) while stimulation of signal **c** causes smooth eye acceleration (Fig. 10F).

In developing the model, we assumed that signal **b** had a value of one during target motion. Fig. 11A shows that the sustained eye velocity of the model in Fig. 9C is consistently smaller than target velocity if signal **b** is given values less than one during target motion. When **b** is 1.0, the output of the model (solid traces) rises quickly and then settles on a value just below target speed. Reducing the value of **b** to 0.75, 0.5, or 0.25 (dashed traces) reduces the initial eye acceleration of the model in the onset of pursuit and prevents simulated eye velocity from matching target velocity, even during prolonged target motion. Figure 11B shows the contrasting effect of reducing the gain of the

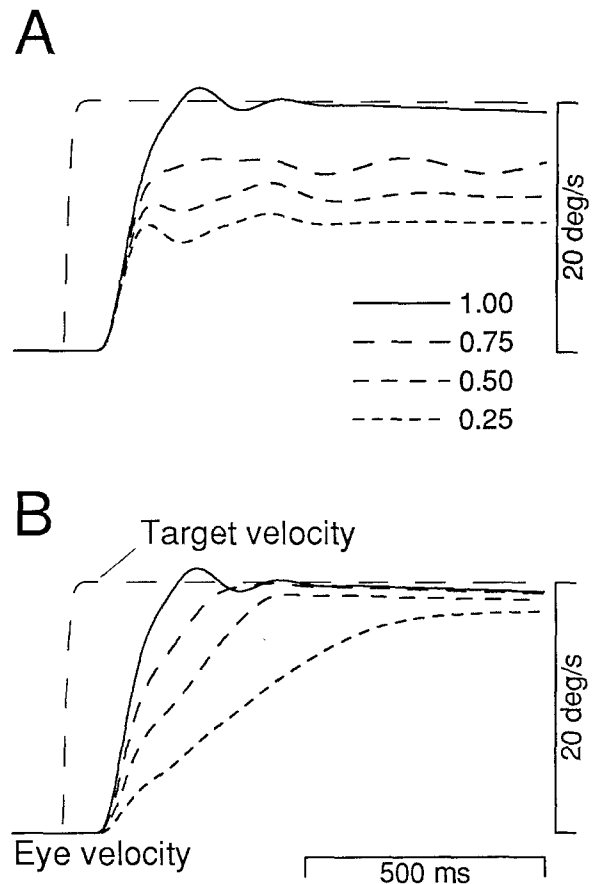


Fig. 11. Effect of lesions at different sites in the model. A: The effect of varying the value assigned to **b** when the target is moving. The solid line shows output of the model when the value of signal **b** was one during pursuit and the other parameters of the model had been optimized to match eye velocity of monkey O in the first 380 ms of pursuit. The other dashed traces show the outputs from the model when the value of signal **b** was set 0.25, 0.50, or 0.75 during target motion. B: Effect of varying the strength of the visual motion inputs. The solid line shows the output of the optimized model for monkey O when the scaling factors on the three visual motion pathways had values of 1. The dashed lines show the output of the same model when the scaling factors at the outputs of all three visual motion pathways had been reduced to 0.25, 0.50, or 0.75.

summed outputs from the three parallel visual motion pathways to 0.75, 0.5 and 0.25 times normal. Although the rate of initial eye acceleration decreased in proportion to the values of the visual gains, eye velocity still climbed toward and ultimately matched or nearly matched target velocity. As long as the positive feedback pathway through  $H_i$  and  $G_{evf}$  was able to properly sustain eye velocity, even grossly reduced visual inputs were sufficient to bring sustained eye velocity nearly up to target velocity.

## Discussion

We have presented a model that reproduces the trajectory of pursuit eye velocity in monkeys and humans at the initiation of pursuit, during sustained tracking of target motion at constant velocity, and at the offset of pursuit. The model depends largely on the dynamics of its visual inputs to generate the dynamics of the initiation and maintenance of pursuit. The visual input consists of three parallel visual pathways that are specialized for transmitting three particular aspects of image motion: image velocity, image acceleration, and a transient input related to a step of target velocity. The structure of these pathways in the model was selected to reflect behavioral demonstrations of three separate components in the first 100 ms of the eye movements evoked by the onset of target motion (Krauzlis and Lisberger, 1994). We want to emphasize that the purpose of our model was to demonstrate the feasibility of the image motion class of model for reproducing the pursuit eye velocity of monkeys on a millisecond time scale. The model was not intended to represent accurately the neural network that provides visual inputs for pursuit in the brain and we have not attempted to use visual inputs that would resemble signals recorded on neurons in visual motion areas such as the middle temporal area (MT). An important next step, for which we have made preliminary reports (Movshon et al., 1990; Lisberger and Movshon, 1991), is to determine how the three visual components in our model are represented in the visual motion pathways.

### *Comparison With Other Models of Pursuit*

There is a general difference between the earlier models of others (Young et al., 1968; Robinson, 1971; Robinson et al., 1986) and the models we have presented here and previously (Morris and Lisberger, 1985; Lisberger et al., 1987; Krauzlis and Lisberger, 1989). The earlier models described the visual pathways as a simple time delay and ascribed most of the dynamics of pursuit to the motor pathways. Our models attribute much of the dynamics of pursuit to the processing of the visual inputs for pursuit, rather than to properties of the output motor pathways. In our model, differential contributions from different visual pathways afford separate control of the dynamics of eye velocity at the onset of pursuit and during pursuit of sustained target motion. The image velocity and

motion onset transient pathways control the initial trajectory of eye velocity for steps of target velocity. The image velocity and image acceleration pathways work together to determine the transition to sustained pursuit and the dynamics of eye velocity during prolonged pursuit of target motion at constant speeds. Although our model does not explicitly represent "early" and "late" components in the initiation of pursuit (e.g. Lisberger and Westbrook, 1985), the "fast" and "slow" dynamics of the motion onset transient and image velocity pathways cause them to generate early and late components, respectively. We customized the parameters of our model only to reproduce the eye velocity evoked by steps of target velocity, but Krauzlis (1991) showed that minor changes in the parameters would allow the same model to reproduce the eye velocity evoked by smooth target acceleration, albeit with a small increase in the difference between the output of the model and the actual eye velocity of the monkeys.

One of the potential drawbacks of the negative feedback configuration of pursuit is that it renders the system prone to unstable oscillations caused by visual feedback with a finite latency. Our model does not eliminate the oscillations caused by visual feedback, but instead uses the parallel visual input pathways to control the amplitude of the oscillations. We designed our model to control rather than to eliminate the oscillations, because oscillations in eye velocity do occur during sustained pursuit tracking both in monkeys (Goldreich et al., 1992) and in humans (Robinson et al., 1986). The oscillations produced by our model are of the same frequency, between 5 and 6 Hz, and amplitude, between 2 and 3°/s, as those observed in the pursuit eye velocity of monkeys. If the parameters are adjusted correctly, then our model emits oscillations at the lower frequency found in the pursuit eye velocity of human subjects (Robinson et al. 1986). Finally, increases in the time delay for visual feedback in the model decrease the frequency of oscillation of simulated eye velocity during pursuit of target motion at constant speed, in good quantitative agreement with the results of comparable experiments in monkeys (Goldreich et al., 1992).

Previous models have maintained stability by driving pursuit eye velocity with an internal, neural reconstruction of target velocity (Young et al., 1968). Robinson (1971) suggested that the internal target velocity signal could be computed by adding an efference copy of eye velocity to afferent signals of image velocity. If the time delay, dynamics, and gain of the internal positive-feedback loop match the time delay, dynamics, and

gain of the external negative-feedback loop, then the visual feedback loop can be counteracted completely and the oscillations caused by visual feedback can be prevented (Robinson et al., 1986). Models that use an internal analog of target velocity as a command for pursuit also can emit oscillations of the correct frequency and amplitude. For example, the model of Robinson et al. (1986) includes an internal negative feedback loop that allows the model to reproduce the 3.8 Hz oscillations observed in the pursuit of human subjects. However, the model of Robinson et al. (1986) does not reproduce the pursuit of monkeys because its frequency of oscillation during sustained pursuit does not decrease as the delay in visual feedback is increased (Goldreich et al., 1992). In contrast, our model can reproduce the pursuit of human subjects.

We conclude that models based on multiple components of visual motion processing currently provide a more accurate description of the primate pursuit system, because these models can reproduce the pursuit eye velocity of both monkeys and humans. As pointed out by Deno et al. (1989), however, it may be possible to modify the model of Robinson et al. (1986) so that it reproduces many of the behaviors of our model. Further, neither the model of Robinson et al. (1986) nor ours is highly constrained by the anatomical organization of the pursuit system or by the kinds of signals recorded in the visual motion areas that provide inputs for pursuit. Thus, it will be necessary to refine all models of pursuit after biological experiments have provided additional constraints on their internal architectures and internal signal processing. Nonetheless, our model provides an essential first step by demonstrating the feasibility of models in which the dynamics of pursuit are determined largely by the temporal dynamics of the visual motion inputs for pursuit.

#### *Relationship Between the Visual Motion Signals in the Model and in the Neural Pathways for Pursuit*

We have presented a model that has three parallel pathways partly because our behavioral experiments (Krauzlis and Lisberger, 1994) provided evidence for three components of visual motion inputs and partly because the three-pathway model provided a good fit to the eye velocity of monkeys during pursuit of steps of target velocity. The three-pathway structure has the practical advantage that steps of target velocity and smooth target accelerations can be used to divide the

visual signals into pieces that can be identified and examined separately (e.g., Krauzlis and Lisberger, 1989; Krauzlis and Lisberger, 1994). However, the three pathways in the model are not meant to imply that the three components of visual motion inputs for pursuit are processed in separate pathways in the brain. For example, the motion onset transient and smooth image acceleration are related to different aspects of image acceleration. They could have been described in the model as a single pathway with a more flexible non-linearity and higher order filters, and they may not be segregated into different pathways or onto different cells at any level of processing in the brain. It is possible that the signals transmitted through all three pathways in our model are combined in the responses of a homogeneous group of cells in the visual motion pathways in the brain. However, the need to adjust the gains of the image velocity and image acceleration pathways independently to reproduce the trial-to-trial and day-to-day variability in pursuit suggests that the different signals may be transmitted over visual motion pathways that are partially segregated.

Two recent experiments have used steps of target velocity and smooth target accelerations to evaluate whether the firing rate of individual neurons is related to image velocity, image acceleration, and/or the motion onset transient (Krauzlis and Lisberger, 1991; Movshon et al., 1990). We have looked for multiple components of visual motion processing in the cerebellum and in extrastriate visual area MT because lesions of these structures cause clear deficits in pursuit eye movements (Newsome et al., 1985; Zee et al., 1981). Recordings from the cerebellum showed that the simple-spike output of the horizontal gaze velocity Purkinje cells in the ventral paraflocculus and flocculus can be accounted for by a combination of the three visual motion components suggested by our model (Krauzlis and Lisberger, 1991), in addition to the well-known signal related to smooth eye velocity (Lisberger and Fuchs, 1978; Stone and Lisberger, 1990). Recordings from MT have shown that the responses of many cells include a transient at the onset of target motion (Movshon et al., 1990). In addition, about half of MT cells show a component of firing rate that can be described as a relation to smooth image acceleration (Lisberger and Movshon, 1991), in addition to the well-known component (e.g. Maunsell and Van Essen, 1983) related to image velocity. However, some MT cells discharge purely in relation to image velocity with no clear motion onset transient and no relationship to image acceleration (Lisberger and Movshon, 1991).



Thus, the responses of MT cells support the possibility mentioned above, that the different signals in our model may be transmitted over neural pathways that are partially segregated.

### *Modeling Eye Velocity Memory for Pursuit*

Several lines of evidence support the postulate that pursuit uses some form of velocity memory. First, behavioral experiments have demonstrated that visual motion inputs cause an acceleration of the eye during pursuit (Lisberger et al., 1981; Lisberger and Westbrook, 1985; Tychsen and Lisberger, 1986; Carl and Gellman, 1987). This implies that ongoing velocity is remembered automatically and that visual inputs cause changes in eye velocity. Second, ongoing eye velocity is automatically sustained when image motion is eliminated by stabilizing the target with respect to the moving eye (Morris and Lisberger, 1987). This requires a mechanism to remember eye velocity in the absence of visual inputs. Third, theoretical analyses argue that eye velocity cannot be driven directly by visual motion inputs because the delays in the processing of visual inputs either promote unstable oscillations or prevent from the system from matching sustained eye speed to target speed (Young et al., 1968). These problems are relieved and stable performance restored in models that employ velocity memory (Lisberger, 1994).

In the final form of our model (Fig. 9C), eye velocity memory is implemented differently from in previous models. Our simpler model (Fig. 1) and previous models of pursuit (Robinson et al., 1986; Krauzlis and Lisberger, 1989) implemented eye velocity memory as a perfect mathematical integrator. Our final model (Fig. 9C) implements velocity memory as a circuit in which perfect integration is achieved by positive feedback around a leaky integrator. The velocity memory is switched between perfect and leaky integration by a switch that is controlled by an extraretinal signal. We implemented velocity memory in this way because the simpler model, with three non-linear visual pathways and perfect integration (Fig. 1), could not reproduce the eye velocity at both the onset and the offset of pursuit. The switch allowed the model to operate in a non-linear mode during pursuit of a moving target and in a linear mode during fixation of a stationary target so that it could reproduce eye velocity at both the onset and offset of pursuit. Other models (e.g. Keller and Kamath, 1976; Zee et al., 1980, 1981) also have used

positive feedback to lengthen the time constant of a leaky integrator. However, both of these models were concerned with a neural integrator that converts commands for eye velocity into the eye position component of motoneuron discharge. For example, the gaze paretic nystagmus seen after lesions of the cerebellar flocculus and ventral paraflocculus has been modeled as a reduction in the gain through a feedback loop that normally improves the performance of a leaky integrator in the brainstem (Zee et al., 1980). The integrator in our model is separate from this velocity-to-position integrator and functions to sustain ongoing eye velocity and to allow visual inputs to serve as commands for eye acceleration.

Our new implementation of velocity memory allowed the model to reproduce several physiological features of the neural pathways for pursuit. Stimulation before the multiplying junction in Fig. 9C caused a change in the eye velocity of the model only if pursuit was already underway. This reproduced the fact that stimulation in the medial superior temporal visual area (MST) (Komatsu and Wurtz, 1989) or the dorso-lateral pontine nucleus (May et al., 1985) produced a clear change in eye velocity only if the monkey was already tracking a moving target. Because the "extraretinal signal" multiplies the eye velocity feedback signal in Fig. 9C, stimulation before the summing junction evokes eye acceleration that is proportional to ongoing eye speed. This agrees with the results of stimulating in area MST in the monkey (Komatsu and Wurtz, 1989). Reduction of the amplitude of the "extraretinal input" to the multiplying junction in Fig. 9C showed how the model might reproduce the deficits in pursuit after lesions of MST (Dursteler and Wurtz, 1988) or the arcuate pursuit area (Lynch, 1987; MacAvoy et al., 1991; Keating et al., 1991, 1993). In monkeys, unilateral lesions of either of these areas cause a deficit in which eye speed does not match target speed for motion toward the side of the lesion but tracking remains normal for target motion away from the side of the lesion. In the model illustrated in Fig. 9C, reductions in the amplitude of the extraretinal input to the multiplying junction caused deficits in the velocity of sustained pursuit even though the visual feedback pathway remained intact with normal gain. Therefore, a version of the model with separate leftward and rightward extraretinal inputs to separate leftward and rightward pursuit systems could reproduce the directional deficits caused by lesions on one side of the cortex.

### *Unresolved Issues Concerning the Generation of Pursuit Eye Movements*

The model presented in this paper highlights several unresolved issues concerning the generation of pursuit eye movements in primates. First, what is the function of the extra-retinal signals found in cortical area MST (Newsome et al., 1988) and the frontal eye fields (MacAvoy et al., 1991)? Our model shows how extraretinal signals could facilitate or gate the transmission of visual information to the smooth ocular motor system without forming part of the basic command for pursuit eye movements. The success of this approach in the model raises the possibility that the extraretinal signals in the brain implement the pursuit "switch" suggested by Grasse and Lisberger (1992) and Goldreich et al. (1992). In contrast, other models of pursuit suggest that the extraretinal signals in the cerebral cortex represent a copy of the eye velocity command used to move the eye and are used to construct a neural correlate of target velocity (Newsome et al., 1988). In fact, it seems likely that extraretinal signals in the cortex have a function more complex than the simple toggle switch of our model but less rigidly defined than the reconstruction of target velocity suggested by other models.

Second, what is the function of eye velocity feedback in pursuit? All models of pursuit now use memory of eye velocity to allow eye speed to match target speed without provoking unstable oscillations in eye velocity. In our earlier model (Krauzlis and Lisberger, 1989; Fig. 1 of this paper) and in the model of Robinson et al. (1986), the velocity memory is provided by an integrator. In the final version of our model (Fig. 9C of this paper), positive feedback of eye velocity is used to lengthen the time constant of a leaky integrator and produce a circuit that mimics the function of a perfect mathematical integrator. The use of eye velocity positive feedback is consistent with the existence of eye velocity signals on horizontal gaze velocity Purkinje cells in the cerebellar flocculus and ventral paraflocculus and with the suggestion that these Purkinje cells subserve a function like that of the positive feedback loop in our model (Lisberger and Fuchs, 1978; Miles et al., 1980; Stone and Lisberger, 1990). However, the gain of the positive feedback through the horizontal gaze velocity Purkinje cells is not known and it is possible that eye velocity feedback through area MST might also provide some of the velocity memory for pursuit (Newsome et al., 1988).

Finally, our model raises important issues by virtue of what it does not describe. The input to the model

is assumed to be image motion, but we have made no attempt to describe how the brain defines the target or quantifies its motion with respect to the eye. The importance and complexity of this issue are underscored by two facts. First, the visual inputs to the brain provide a highly non-linear representation of image motion, but the input to the three visual motion pathways in our model is a signal that is strictly proportional to image velocity. Future models of pursuit should strive to correct this glaring deficiency. Second, most objects of ocular pursuit appear in rich three-dimensional environments that are very different from the featureless backgrounds used in our experiments. How the brain parses the motion contained in such complicated visual environments, how it determines which portions of the motion to use as inputs for pursuit eye movements, and how that motion is represented in the brain pathways for pursuit remain as vital questions for future biological experiments and computer simulations.

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